

Supplemental Data

The Immigrant Birthweight Paradox in an Urban Cohort: *Role of Immigrant Enclaves and Ambient Air Pollution*

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Figure S1. Conceptual model of the relationship between maternal residence in immigrant enclaves, ambient air pollution sources, and birthweight-for-gestational-age z-score

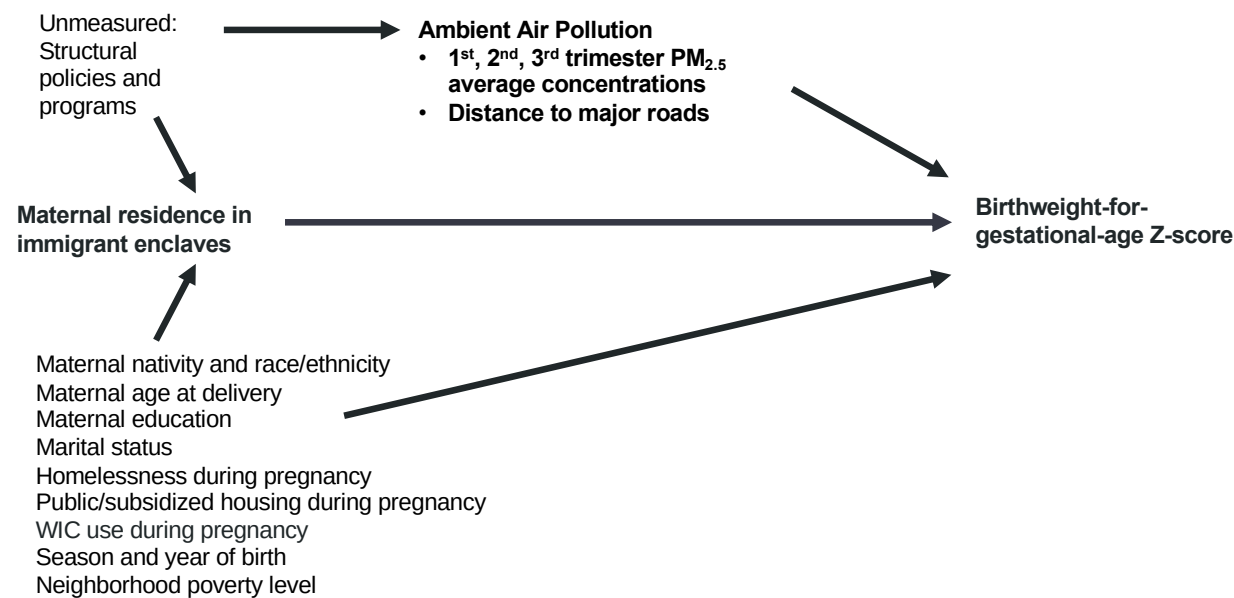


Table S1. American Community Survey variables used in confirmatory factor analysis of immigrant enclave, Boston-based Children’s HealthWatch cohort, 2006-2015 (N=2,514)

	Correlation of item with total score*	Item whole correlation for item against the scale without item	Mean	Standard Deviation
ACS 2006-2010 five-year data				
Immigrant enclave score Cronbach’s alpha: 0.939 (95% CI: 0.928 – 0.949) Average interitem correlation: 0.835	NA	NA	0.488	0.134
Percent non-English only Households	0.823	0.804	0.590	0.144
Percent Foreign Born Population	0.965	0.928	0.495	0.139
Percent Non-U.S.-Citizen Population	0.937	0.887	0.379	0.145
ACS 2011-2015 five-year data				
Immigrant enclave score Cronbach’s alpha: 0.942 (95% CI: 0.931 - 0.951) Average interitem correlation: 0.850	NA	NA	0.493	0.136
Percent non-English only Households	0.847	0.829	0.598	0.151
Percent Foreign Born Population	0.963	0.929	0.505	0.138
Percent Non-US-Citizen Population	0.937	0.892	0.378	0.139

* Item whole correlation corrected for item overlap and scale reliability

Table S2. Spearman correlation coefficients of immigrant enclave score components and environmental factors, Boston-based Children's HealthWatch cohort, 2006-2015 (N=2,514)

	Immigrant Enclave Score	Linguistically isolated households (%)	Foreign-born Populations (%)	Non-US citizen Populations (%)	First Trimester PM_{2.5} (µg/m³)	Second Trimester PM_{2.5} (µg/m³)	Third Trimester PM_{2.5} (µg/m³)	Proximity to major roads (m)	Population Living in Poverty (%)
Geographic scale	Tract	Tract	Tract	Tract	1-km ² grid	1-km ² grid	1-km ² grid	Euclidean horizontal distance	Tract
Immigrant Enclave Score	1.000	0.747	0.995	0.907	-0.009	-0.030	-0.025	-0.018	0.014
Linguistically isolated households (%)	0.747	1.000	0.735	0.688	-0.042	-0.088	-0.062	-0.046	0.290
Foreign-born Populations (%)	0.995	0.735	1.000	0.881	-0.026	-0.050	-0.045	-0.012	0.003
Non-US citizen Populations (%)	0.907	0.688	0.881	1.000	0.009	0.003	-0.002	-0.069	0.081
First Trimester PM_{2.5} (µg/m³)	-0.009	-0.042	-0.026	0.009	1.000	0.172	0.511	0.020	0.014
Second Trimester PM_{2.5} (µg/m³)	-0.030	-0.088	-0.050	0.003	0.172	1.000	0.158	-0.002	0.011
Third Trimester PM_{2.5} (µg/m³)	-0.025	-0.062	-0.045	-0.002	0.511	0.158	1.000	0.003	0.029
Proximity to major roads (m)	-0.018	-0.046	-0.012	-0.069	0.020	-0.002	0.003	1.000	-0.081
Population Living in Poverty (%)	0.014	0.290	0.003	0.081	0.014	0.011	0.029	-0.081	1.000

Table S3. Sociodemographic, neighborhood, and birthweight characteristics by immigrant enclave score percentiles, Boston-based Children's HealthWatch cohort, 2006-2015 (N=2,514)

	Immigrant Enclave Score by Percentile of the Statewide Distribution			
	Total (N=2,514) n (%)	0 to 79 th Percentile (N= 730) n (%)	80 to 89 th Percentile (N= 611) n (%)	90 to 100 th Percentile (N=1,173) n (%)
Maternal Characteristics				
Nativity status				
US-born	1,213 (48.2%)	444 (60.8%)	309 (50.6%)	460 (39.2%)
Foreign-Born	1,301 (51.8%)	286 (39.2%)	302 (49.4%)	713 (60.8%)
Race/ethnicity				
Latina	972 (38.7%)	210 (28.8%)	207 (33.9%)	555 (47.3%)
Black	1,240 (49.3%)	403 (55.2%)	330 (54.0%)	507 (43.2%)
White	179 (7.1%)	88 (12.1%)	39 (6.4%)	52 (4.4%)
Other	123 (4.9%)	29 (4.0%)	35 (5.7%)	59 (5.0%)
Language of interview				
English	2,047 (81.4%)	664 (91.0%)	547 (89.5%)	836 (71.3%)
Spanish	467 (18.6%)	66 (9.0%)	64 (10.5%)	337 (28.7%)
Length of time in the U.S.				
US-born	1,213 (48.2%)	444 (60.8%)	309 (50.6%)	460 (39.2%)
<5 years in the US	144 (5.7%)	24 (3.3%)	27 (4.4%)	93 (7.9%)
5-10 years in the US	565 (22.5%)	118 (16.2%)	130 (21.3%)	317 (27.0%)
11+ years in the US	506 (20.1%)	115 (15.8%)	120 (19.6%)	271 (23.1%)
Unknown	86 (3.4%)	29 (4.0%)	25 (4.1%)	32 (2.7%)
Top countries/regions of origin				
United States – 50 states	1,138 (45.3%)	416 (57.0%)	291 (47.6%)	431 (36.7%)
United States – Puerto Rico	75 (3.0%)	28 (3.8%)	18 (2.9%)	29 (2.5%)
El Salvador	233 (9.3%)	8 (1.1%)	28 (4.6%)	197 (16.8%)
Haiti	188 (7.5%)	48 (6.6%)	60 (9.8%)	80 (6.8%)
Cape Verde	141 (5.6%)	28 (3.8%)	26 (4.3%)	87 (7.4%)
Dominican Republic	136 (5.4%)	44 (6.0%)	34 (5.6%)	58 (4.9%)
Educational status				
Never/Elementary/Some High school	571 (22.7%)	108 (14.8%)	113 (18.5%)	350 (29.8%)
High School	833 (33.1%)	238 (32.6%)	200 (32.7%)	395 (33.7%)
Tech School/College Graduate/Masters	1,110 (44.2%)	384 (52.6%)	298 (48.8%)	428 (36.5%)
Marital status				
Married/Partner/Cohabitation	1,007 (40.1%)	278 (38.1%)	243 (39.8%)	486 (41.4%)
Single/Never-married	1,198 (47.7%)	378 (51.8%)	299 (48.9%)	521 (44.4%)
Separate/Divorced/Widowed	309 (12.3%)	74 (10.1%)	69 (11.3%)	166 (14.2%)
Age at delivery				
Mean (SD)	27.2 (6.31)	27.5 (6.59)	27.3 (6.32)	27.0 (6.12)
WIC use during pregnancy				
No	354 (14.1%)	143 (19.6%)	100 (16.4%)	111 (9.5%)
Yes	2,160 (85.9%)	587 (80.4%)	511 (83.6%)	1,062 (90.5%)
Public/subsidized housing during pregnancy				
No	1,736 (69.1%)	479 (65.6%)	398 (65.1%)	859 (73.2%)
Yes	778 (30.9%)	251 (34.4%)	213 (34.9%)	314 (26.8%)
Homelessness anytime during pregnancy				
No	2,293 (91.2%)	656 (89.9%)	546 (89.4%)	1,091 (93.0%)
Yes	221 (8.8%)	74 (10.1%)	65 (10.6%)	82 (7.0%)
Neighborhood Social Factors (Census tract-level)				
Immigrant enclave score tract-level				
Mean (SD)	1.24 (0.694)	0.433 (0.529)	1.16 (0.113)	1.79 (0.369)
Linguistically isolated households[^] (%)				

Mean (SD)	0.456 (0.153)	0.329 (0.122)	0.420 (0.0969)	0.554 (0.124)
Foreign-born Populations (%)				
Mean (SD)	0.324 (0.124)	0.185 (0.0588)	0.292 (0.0237)	0.426 (0.0861)
Non-US citizen Populations (%)				
Mean (SD)	0.180 (0.116)	0.0843 (0.0365)	0.143 (0.0342)	0.259 (0.121)
Percent in poverty tract-level[†] (%)				
Mean (SD)	0.211 (0.114)	0.212 (0.137)	0.213 (0.117)	0.209 (0.0963)
Child's Characteristics				
Birthweight-for-gestational age z-score				
Mean (SD)	-0.218 (0.913)	-0.236 (0.932)	-0.266 (0.922)	-0.182 (0.897)
Birthweight (Grams)				
Mean (SD)	3,310 (486)	3,300 (495)	3,290 (494)	3,330 (475)
Birthweight-for-gestational age percentile (%)				
Mean (SD)	41.5 (26.9)	41.0 (27.3)	40.1 (27.2)	42.4 (26.5)
Gender				
Male	1,364 (54.3%)	387 (53.0%)	339 (55.5%)	638 (54.4%)
Female	1,150 (45.7%)	343 (47.0%)	272 (44.5%)	535 (45.6%)

Note: The immigrant enclave score was estimated using a confirmatory factor analysis of American Community Survey estimates for the percentages of linguistically isolated households, foreign-born populations, and non-US citizen populations for each census tract in Massachusetts. We then transformed the z-score into percentiles (0-100) and divided our CHW study cohort into three ordered score percentiles (i.e. 0-79th, 80-8th, and 90-100th). As shown, the majority of mothers in our cohort lived in areas with very high immigrant enclave scores (i.e. in the 80+ percentile).

[^] Linguistically-isolated households: No one age 14+ years old speaks English-only or speaks English "very well".

[†] Among whom poverty status is determined.

Black, White, and Other race categories refer to mothers who were not Hispanic, Latina, or Spanish.

Abbreviations: SD = Standard deviation.

Table S4. Adjusted mean differences in birthweight-for-gestational-age z-scores in stratified models – US-born and foreign-born mothers, Boston-based Children’s HealthWatch cohort, 2006-2015

	US-born Mothers (N=1,213)		Foreign-born Mothers (N=1,301)	
	Beta	95% CI	Beta	95% CI
Intercept	-30.56	(-105.81,44.69)	-45.75	(-117.94,26.44)
Immigrant Enclave Score	-0.052	(-0.130,0.027)	0.090	(0.007,0.172)
Maternal Race/Ethnicity (ref: Black)				
Latina	0.013	(-0.106,0.131)	0.013	(-0.105,0.132)
White	0.319	(0.133,0.506)	0.341	(0.013,0.669)
Other	-0.078	(-0.289,0.133)	-0.002	(-0.231,0.227)
PM_{2.5} concentrations (per 10 µg/m ³)				
First Trimester Average	0.389	(-0.043,0.822)	0.079	(-0.346,0.504)
Second Trimester Average	-0.209	(-0.644,0.227)	0.108	(-0.299,0.514)
Third Trimester Average	-0.111	(-0.552,0.329)	-0.033	(-0.439,0.372)
Proximity to major roadways (per 10 m)				
Live within 100 m of road	-0.012	(-0.033, 0.010)	-0.025	(-0.046, -0.004)
Live beyond 100 m of road	-0.0005	(-0.004, 0.003)	-0.006	(-0.011, -0.001)
Covariates				
Age at delivery (centered, years)	0.011	(0.002,0.019)	0.008	(-0.001,0.017)
WIC use during pregnancy	0.045	(-0.107,0.196)	0.057	(-0.109,0.224)
Homelessness during pregnancy	-0.209	(-0.371,-0.048)	-0.009	(-0.235,0.216)
Public/subsidized housing during pregnancy	-0.008	(-0.114,0.097)	0.131 [^]	(-0.007,0.269)
Population in poverty (centered, %)	0.187	(-0.228,0.602)	0.093	(-0.435,0.622)

Note: **Bold** indicates statistical significance ($p < 0.05$). [^] = marginal significance ($p = 0.05 - 0.10$).

Models adjusted for year and season of child’s birth.

Black, White, and Other race categories refer to mothers who were not Hispanic, Latina, or Spanish.

Abbreviations: Beta = Estimated mean difference in z-scores; 95% CL = 95% Confidence limit.

Table S5. Adjusted mean differences in birthweight-for-gestational-age z-scores comparing final model with models (A) restricting to dyads with an EMR residential address at child's birth (B) and mother-child dyads from the Boston Metropolitan area (C), Boston-based Children's HealthWatch cohort, 2006-2015

	Model A. Final model (N=2,514)		Model B. Dyads with EMR address at child's birth (N=1,788)		Model C. From Boston Metropolitan area (N=2,475)	
	Beta	95% CL	Beta	95% CL	Beta	95% CL
Intercept	-36.84	(-88.38,14.71)	-32.76	(-92.31,26.79)	-33.64	(-85.87,18.59)
Immigrant Enclave Score	-0.055	(-0.131,0.022)	-0.047	(-0.145,0.051)	-0.064	(-0.143,0.014)
Foreign-born status	0.015	(-0.137,0.168)	-0.041	(-0.229,0.146)	0.022	(-0.138,0.181)
Maternal Race/Ethnicity (ref: Black)						
Latina	0.011	(-0.072,0.094)	0.003	(-0.095,0.102)	0.009	(-0.074,0.092)
White	0.334	(0.176,0.492)	0.330	(0.124,0.537)	0.309	(0.147,0.472)
Other	-0.045	(-0.198,0.109)	-0.052	(-0.237,0.133)	-0.059	(-0.218,0.100)
PM_{2.5} concentrations (per 10 µg/m ³)						
First Trimester Average	0.230	(-0.069,0.529)	0.289	(-0.059,0.637)	0.236	(-0.067,0.538)
Second Trimester Average	-0.054	(-0.349,0.241)	-0.048	(-0.393,0.296)	-0.068	(-0.367,0.231)
Third Trimester Average	-0.067	(-0.036,0.228)	-0.349	(-0.676,-0.022)	-0.085	(-0.383,0.213)
Proximity to major roads (per 10 m)						
Live within 100 m of road	-0.018	(-0.033, -0.003)	-0.016 [^]	(-0.034, 0.002)	-0.017	(-0.032, -0.002)
Live beyond 100 m of road	-0.003 [^]	(-0.006,-0.00003)	-0.002	(-0.005,0.002)	-0.003	(-0.006, -0.00003)
Covariates						
Age at delivery (centered, years)	0.009	(0.003,0.015)	0.012	(0.005,0.019)	0.009	(0.003,0.015)
WIC use during pregnancy	0.044	(-0.067,0.156)	0.096	(-0.044,0.235)	0.043	(-0.071,0.157)
Homelessness during pregnancy	-0.132	(-0.262,-0.001)	-0.193	(-0.346,-0.04)	-0.132	(-0.263,-0.002)
Public/subsidized housing during pregnancy	0.061	(-0.023,0.145)	0.067	(-0.034,0.169)	0.062	(-0.023,0.146)
Population in poverty (centered, %)	0.119	(-0.205,0.443)	-0.134	(-0.517,0.25)	0.075	(-0.252,0.402)
Interaction: Immigrant Enclave Score x Foreign-born	0.134	(0.027,0.241)	0.138	(0.007,0.269)	0.133	(0.022,0.244)

Note: **Bold** indicates statistical significance ($p < 0.05$). [^] = marginal significance ($p = 0.05 - 0.10$).

Models adjusted for year and season of child's birth.

Black, White, and Other race categories refer to mothers who were not Hispanic, Latina, or Spanish.

Abbreviations: Beta = Estimated mean difference in z-scores; 95% CL = 95% Confidence limit.

Table S6. Adjusted mean differences in birthweight-for-gestational-age z-scores comparing final model with a model restricted to children with birthweights between 2,500 and 4,000 grams (i.e. excluding low birthweight and macrosomia infants), Boston-based Children’s HealthWatch cohort, 2006-2015

	Final model (N=2,514)		Children with birthweight 2,500-4,000 grams (N=2,208)	
	Beta	95% CL	Beta	95% CL
Intercept	-36.84	(-88.38,14.71)	-18.14	(-57.79,21.50)
Immigrant Enclave Score	-0.055	(-0.131,0.022)	-0.028	(-0.09,0.033)
Foreign-born status	0.015	(-0.137,0.168)	-0.029	(-0.147,0.089)
Maternal Race/Ethnicity (ref: Black)				
Latina	0.011	(-0.072,0.094)	-0.016	(-0.078,0.047)
White	0.334	(0.176,0.492)	0.163	(0.044,0.282)
Other	-0.045	(-0.198,0.109)	-0.040	(-0.162,0.083)
PM_{2.5} concentrations (per 10 µg/m ³)				
First Trimester Average	0.230	(-0.069,0.529)	0.048	(-0.188,0.284)
Second Trimester Average	-0.054	(-0.349,0.241)	-0.085	(-0.317,0.147)
Third Trimester Average	-0.067	(-0.036,0.228)	-0.023	(-0.254,0.207)
Proximity to major roads (per 10 m)				
Live within 100 m of road	-0.018	(-0.033, -0.003)	-0.012	(-0.024,-0.0003)
Live beyond 100 m of road	-0.003 [^]	(-0.006,-0.00003)	-0.003	(-0.005,-0.0002)
Covariates				
Age at delivery (centered, years)	0.009	(0.003,0.015)	0.007	(0.002,0.012)
WIC use during pregnancy	0.044	(-0.067,0.156)	0.061	(-0.024,0.146)
Homelessness during pregnancy	-0.132	(-0.262,-0.001)	-0.068	(-0.167,0.031)
Public/subsidized housing during pregnancy	0.061	(-0.023,0.145)	-0.021	(-0.085,0.042)
Population in poverty (centered, %)	0.119	(-0.205,0.443)	0.059	(-0.195,0.312)
Interaction: Immigrant Enclave Score x Foreign-born	0.134[†]	(0.027,0.241)	0.085^{††}	(0.002,0.168)

Note: **Bold** indicates statistical significance (p < 0.05). [^] = marginal significance (p = 0.05 – 0.10). Models adjusted for year and season of child’s birth. [†]p-interaction for final model = 0.014; ^{††} p-interaction for sensitivity model = 0.044. Black, White, and Other race categories refer to mothers who were not Hispanic, Latina, or Spanish. Abbreviations: Beta = Estimated mean difference in z-scores; 95% CL = 95% Confidence limit.

Table S7. Adjusted mean differences in birthweight-for-gestational-age z-scores by child sex, Boston-based Children's HealthWatch cohort, 2006-2015

	Final model (N=2,514)		Male Children (N=1,364)		Female Children (N=1,150)	
	Beta	95% CL	Beta	95% CL	Beta	95% CL
Intercept	-36.84	(-88.38,14.71)	-51.32	(-118.78,16.14)	-3.06	(-84.37,78.26)
Immigrant Enclave Score	-0.055	(-0.131,0.022)	-0.027	(-0.132,0.077)	-0.076	(-0.189,0.037)
Foreign-born status	0.015	(-0.137,0.168)	-0.033	(-0.232,0.166)	0.056	(-0.181,0.293)
Maternal Race/Ethnicity (ref: Black)						
Latina	0.011	(-0.072,0.094)	0.010	(-0.096,0.115)	-0.006	(-0.138,0.127)
White	0.334	(0.176,0.492)	0.370	(0.150,0.590)	0.287	(0.055,0.519)
Other	-0.045	(-0.198,0.109)	0.007	(-0.228,0.242)	-0.063	(-0.274,0.147)
PM_{2.5} concentrations (per 10 µg/m ³)						
First Trimester Average	0.230	(-0.069,0.529)	0.437	(0.047,0.827)	-0.064	(-0.533,0.405)
Second Trimester Average	-0.054	(-0.349,0.241)	-0.159	(-0.560,0.243)	0.004	(-0.439,0.446)
Third Trimester Average	-0.067	(-0.036,0.228)	-0.086	(-0.464,0.292)	-0.060	(-0.532,0.412)
Proximity to major roads (per 10 m)						
Live within 100 m of road	-0.018	(-0.033, -0.003)	-0.012	(-0.032,0.009)	-0.025	(-0.048,-0.002)
Live beyond 100 m of road	-0.003 [^]	(-0.006,-0.00003)	-0.003	(-0.007,0.001)	-0.003	(-0.008,0.001)
Covariates						
Age at delivery (centered, years)	0.009	(0.003,0.015)	0.008 [^]	(0.000,0.015)	0.011	(0.001,0.021)
WIC use during pregnancy	0.044	(-0.067,0.156)	-0.002	(-0.146,0.143)	0.099	(-0.08,0.278)
Homelessness during pregnancy	-0.132	(-0.262,-0.001)	-0.006	(-0.164,0.153)	-0.281	(-0.494,-0.068)
Public/subsidized housing during pregnancy	0.061	(-0.023,0.145)	0.023	(-0.084,0.131)	0.114 [^]	(-0.021,0.249)
Population in poverty (centered, %)	0.119	(-0.205,0.443)	0.194	(-0.212,0.600)	-0.016	(-0.55,0.518)
Interaction: Immigrant Enclave Score x Foreign-born	0.134	(0.027,0.241)	0.119 [^]	(-0.022,0.26)	0.157 [^]	(-0.005,0.318)

Note: **Bold** indicates statistical significance ($p < 0.05$). [^] = marginal significance ($p = 0.05 - 0.10$).

Models adjusted for year and season of child's birth. [†] p -interaction = 0.014.

Black, White, and Other race categories refer to mothers who were not Hispanic, Latina, or Spanish.

Abbreviations: Beta = Estimated mean difference in z-scores; 95% CL = 95% Confidence limit.